

FMPT-7000 Smart Gauge / Absolute Pressure Transmitter

The FMPT-7000 is a high-performance Smart Gauge and Absolute Pressure Transmitter engineered for demanding industrial measurement and control applications. Built on a rugged, all-welded stainless steel sensing platform, it delivers dependable accuracy across an ultra-wide measurement span — 0.1 bar to 400 bar — with a 100:1 turn-down ratio, making it a versatile choice for process, utility, and offshore pressure measurement duties.

General Features

- Gauge (GP) and Absolute (AP) versions on a common platform: GP 0.4–400 bar, AP 0.1–100 bar.
- Best-in-class accuracy: $\pm 0.075\%$ URL (GP) / $\pm 0.1\%$ URL (AP), with $\pm 0.2\%$ SPAN / 10-year long-term stability.
- HART communication with 4-20 mA DC output; optional integrated Bluetooth (iB) for wireless configuration.
- Global hazardous-area certifications: ATEX, IECEx, CSA, PCEC, NEPSI, EAC.
- IP67 rated; ceramic measuring diaphragm available for low AP ranges (0.1 & 0.2 bar).



Key Specifications

Measurement	Gauge (GP) & Absolute (AP)
Range	GP 0.4-400 bar; AP 0.1-100 bar
Accuracy	GP $\pm 0.075\%$; AP $\pm 0.1\%$ URL
Turn-Down	100:1
Output	4-20 mA HART, optional Bluetooth
Certification	ATEX, IECEx, CSA, PCEC, NEPSI, EAC
IP Rating	IP67

Industries & Applications

- Oil & Gas — wellhead, pipeline, and refinery gauge/absolute pressure measurement.
- Power Generation — boiler drum, feedwater, and steam header pressure monitoring.
- Chemical & Petrochemical — reactor and vessel pressure with Hastelloy / ceramic wetted-part options.
- Water & Wastewater — pump discharge and network pressure monitoring.
- Marine & Offshore — DNV, LR, ABS, BV, KR, ClassNK, RS certified for vessel and FPSO applications.

General Specifications — FMPT-7000

Parameter	Specification
Measurement Type	Gauge Pressure (GP) and Absolute Pressure (AP)
Nominal Range	GP: 0.4 bar to 400 bar (6 ranges) AP: 0.1 bar to 100 bar (7 ranges)
Turn-Down Ratio	100:1
Accuracy (Linear)	GP: $\pm 0.075\%$ to $\pm 0.1\%$ URL AP: $\pm 0.1\%$ to $\pm 0.2\%$ URL
Long-Term Stability	$\pm 0.2\%$ Span / 10 years
Output Signal	4-20 mA DC with HART; optional integrated Bluetooth (/B)
Sensor Structure	Single flange diaphragm seal; ceramic diaphragm for AP 0.1/0.2 bar
Supply Voltage	18.3-44 V DC (HART); 18.3-30 V DC (intrinsically safe HART)
Power Consumption	500 mW max @ 24 V DC, 20.8 mA
Load Resistance	0-1476 Ohm (operating); 250-800 Ohm (HART communication)
Damping (Electronic)	0-100 s configurable; sensor module damping ≤ 0.2 s
Response Time	100 ms max
IP Rating	IP67
Weight	Approx. 1.56 kg (excluding mounting bracket and process accessories)

Measurement Range Specifications

Type	Nominal Range	Min. Range	Lower (LRL)	Upper (URL)	Overload
GP	0.4 bar	20 mbar	-0.4 bar	0.4 bar	10 bar
GP	2.5 bar	125 mbar	-1 bar	2.5 bar	40 bar
GP	10 bar	0.5 bar	-1 bar	10 bar	60 bar
GP	30 bar	1.5 bar	-1 bar	30 bar	150 bar
GP	100 bar	5 bar	-1 bar	100 bar	200 bar
GP	400 bar	50 bar	-1 bar	400 bar	800 bar
AP	0.1 bar	50 mbar	0 bar	0.1 bar	6 bar
AP	0.2 bar	100 mbar	0 bar	0.2 bar	10 bar
AP	0.4 bar	0.2 bar	0 bar	0.4 bar	10 bar
AP	2.5 bar	0.5 bar	0 bar	2.5 bar	40 bar
AP	10 bar	2 bar	0 bar	10 bar	60 bar
AP	30 bar	3 bar	0 bar	30 bar	150 bar
AP	100 bar	10 bar	0 bar	100 bar	200 bar

Note: LRV/URV may be set within measurement range limits. When AP range is 0.1 or 0.2 bar, the measuring diaphragm is ceramic.

Accuracy — Linear Output

Type	Condition	Range	Accuracy
GP	TD ≤ 5	0.4 bar	±0.1%
GP	TD ≤ 5	2.5, 10, 30, 100, 400 bar	±0.075%
GP	TD > 5	0.4 bar	±(0.025 + 0.015 × TD)%
GP	TD > 5	2.5, 10, 30, 100, 400 bar	±(0.0025 + 0.0145 × TD)%
AP	TD ≤ 5	0.1, 0.2, 0.4 bar	±0.2%
AP	TD ≤ 5	2.5, 10, 30, 100 bar	±0.1%
AP	TD > 5	0.1, 0.2, 0.4 bar	±(0.025 + 0.035 × TD)%
AP	TD > 5	2.5, 10, 30, 100 bar	±(0.025 + 0.015 × TD)%

Note: TD (Turn-Down) = URL (maximum/factory-calibrated range) / Current SPAN (|URV-LRV|). Reference: 20°C ±5°C, calibrated from zero.

Ambient Temperature Effects

Type	Effect	Applicable Range
GP	±(0.075 + 0.0375 × TD)% per 10°C of SPAN	0.4, 2.5, 10, 30, 100, 400 bar
AP	±(0.125 + 0.075 × TD)% per 10°C of SPAN	0.1, 0.2, 0.4 bar
AP	±(0.115 + 0.065 × TD)% per 10°C of SPAN	2.5, 10, 30, 100 bar

Environmental Conditions

Parameter	Specification
Ambient Temperature	Without LCD: -50 to +85°C; With LCD: -40 to +70°C
Storage Temperature	Without LCD: -50 to +100°C; With LCD: -40 to +85°C
Sensor Operating Temperature	-40 to +105°C
Medium Temp. (Silicone Fill)	-40 to +200°C (standard); -55 to +200°C (low-temp silicone)
Medium Temp. (Fluorocarbon)	-20 to +110°C (Fluorocarbon 1); -55 to +85°C (Fluorocarbon 2, oxygen service)
Ambient Humidity	5% RH to 100% RH @ 40°C
Voltage Effect	For supply voltage within 18.3-44 V DC, zero/span change ≤0.005% URL/V
Vibration Effects	< 0.1% Span per GB/T18271.3 / IEC 61298-3
Mounting Position Error	< 4 mbar (correctable by PV = 0 reset at installation)

EMC Performance

Test	Standard	Test Condition	Level
Radiated Interference (Housing)	GB/T 9254.1 / CISPR 32	30 MHz - 1000 MHz	Qualified
Conducted Interference (DC Port)	GB/T 9254.1 / CISPR 32	0.15 MHz - 30 MHz	Qualified
Electrostatic Discharge Immunity	GB/T 17626.2 / IEC 61000-4-2	8 kV contact / 15 kV air	B
Radiated RF/EM Field Immunity	GB/T 17626.3 / IEC 61000-4-3	10 V/m (80 MHz - 1 GHz)	A
Power Frequency Magnetic Field	GB/T 17626.8 / IEC 61000-4-8	30 A/m	B
Electrical Fast Transient / Burst	GB/T 17626.4 / IEC 61000-4-4	4 kV (5/50 ns, 100 kHz)	B
Surge Immunity	GB/T 17626.5 / IEC 61000-4-5	2 kV (L-L); 4 kV (L-G)	B
RF Conducted Disturbance Immunity	GB/T 17626.6 / IEC 61000-4-6	3 V/m (150 kHz - 80 MHz)	A

Note: Level A: normal operation within limits. Level B: temporary degradation with auto-recovery and no data loss.

Power Supply, Load & Time Index

Parameter	Specification
Power Supply Voltage	HART: 18.3-44 V DC Intrinsically safe HART: 18.3-30 V DC
Load Resistance	0-1476 Ohm (operation); 250-800 Ohm (HART communication)
Transmission Distance	< 1000 m
Power Consumption	≤ 500 mW @ 24 V DC, 20.8 mA
Damping (Electronic)	0-100 s configurable
Sensor Module Damping	≤ 0.2 s
Turn-On Time	≤ 6 s
Factory Reset Time	≤ 31 s
Response Time	≤ 100 ms

Hazardous Area Certifications

Standard	Classification Mark	Certifying Body
Ex Intrinsically Safe	Ex ia IIC T4 Ga	PCEC (China)
Ex Flameproof	Ex db IIC T6 Gb	PCEC (China)
Ex Dust Ignition-proof	Ex tb IIC T85°C Db	NEPSI (China)
ATEX Intrinsically Safe	II 1G Ex ia IIC T4 Ga	CSA / ATEX
ATEX Flameproof	II 2G Ex db IIC T6 Gb	CSA / ATEX
ATEX Dust Explosion-proof	II 2D Ex tb IIIC T80°C Db	CSA / ATEX
IECEX Intrinsically Safe	Ex ia IIC T4 Ga	IECEX / CSA
IECEX Flameproof	Ex db IIC T6 Gb	IECEX / CSA
IECEX Dust Explosion-proof	Ex tb IIIC T80°C Db	IECEX / CSA
NA Intrinsically Safe	Class I Div. 1 Grp A-D T4; Zone 0 AEx ia IIC T4 Ga	CSA

Standard	Classification Mark	Certifying Body
NA Flameproof	Class I Div. 1 Grp A-D T6; Zone 1 AEx db IIC T6 Gb	CSA
EAC Explosion-proof	1Ex db IIC T6 Gb; 0Ex ia IIC T4 Ga	EAC (Russia/CIS)

Note: Some certifications are in process. Contact Forbes Marshall for current certificate status and copies.

Marine & Classification Society Approvals

Society	Full Name	Certificate No.
CCS	China Classification Society	TJ23PTB00014
DNV	Det Norske Veritas	TAA00000Y1
BV	Bureau Veritas	77759/A0 BV
ABS	American Bureau of Shipping	24-2536413-PDA
LR	Lloyd's Register	LR2520052TA
KR	Korean Register	NAJ50160-AE001
NK	Nippon Kaiji Kyokai (ClassNK)	TA24376M
RS	Russian Maritime Register of Shipping	24.44.01.01398.266

Hazardous Area — Order Code (Model Code Position 1)

Code	Protection Type	Certifying Body
0	Non-hazardous area	—
1	Flameproof	PCEC (China)
2	Intrinsically Safe	PCEC (China)
3	Dust Explosion-proof	NEPSI (China)
4	Flameproof + Intrinsically Safe	PCEC / NEPSI (China)
5	Flameproof + Dust Explosion-proof + Intrinsically Safe	PCEC / NEPSI (China)
A	Flameproof	CSA
B	Intrinsically Safe	CSA
C	Flameproof + Intrinsically Safe	CSA
E	Flameproof	ATEX
F	Intrinsically Safe	ATEX
G	Flameproof + Intrinsically Safe	ATEX
J	Flameproof	IECEX
K	Intrinsically Safe	IECEX
L	Flameproof + Intrinsically Safe	IECEX
M	Explosion-proof	EAC (Russia/CIS)

Wetted Parts & Sensor Structure

Sensor structure: single flange diaphragm seal (Code T).

Code	Diaphragm	Process Connection Thread	Notes
A	316L SS	316L SS	Standard process service
B	HC-276 (Hastelloy)	316L SS	Highly corrosive media
C	Tantalum	316L SS	Extreme corrosion resistance (extended lead time)
E	Ceramic	316L SS	Only with G-code process connection; required for AP 0.1/0.2 bar

Output Signal & Communication Options

Option	Description
4-20 mA DC, HART	Standard 2-wire loop powered; default HART5 (HART7 selectable via /H7)
4-20 mA DC, HART + Bluetooth	Integrated BLE module for wireless configuration and monitoring via mobile app
Damping (/ST)	Adjustable 0-100 s, contract to specify value
Fault Alarm High (/WH)	20.8 mA (HART5) / 21 mA (HART7)
Fault Alarm Low (/WL)	3.8 mA (HART5, default) / 3.6 mA (HART7)

Process Connection Options

Code	Specification	Vent Hole
M	M20 x 1.5 Male	Phi 3 mm
G	G1/2 Male	Phi 3 mm (only option for ceramic diaphragm, port Phi 10 mm)
A	1/2-14 NPT Male	Phi 6 mm
N	1/2-14 NPT Female	Phi 14 mm
H	Flange (non-diaphragm)	Per /H01, /H02, /H08 raised-face flange options

Case, Connector & Mounting Options

Code	Description
S	316L Stainless steel case, dual outlet ports (F) M20 x 1.5
U	316L Stainless steel case, dual outlet ports (F) 1/2 NPT
P	Aluminium alloy (polyester resin coated), dual outlet ports (F) M20 x 1.5
N	Aluminium alloy (polyester resin coated), dual outlet ports (F) 1/2 NPT
Display N / L	N = without LCD; L = with LCD (both -40 to +70°C explosion-proof rating)
Mounting (/G7, /G8)	L-shaped bracket, 316 SS (/G7) or Q235 carbon steel (/G8); pipe- or plate-mounted

Waterproof / Explosion-proof Connector (Model Code Position 4)

Code	Description	Material	Cable Dia.	IP Rating
0	Without cable gland	—	—	—
A	Single-ended, flameproof metal plug	316 SS	—	—
1	M20x1.5 waterproof connector, with plug	PVC	6-12 mm	IP67
2	Non-flameproof adapter M20x1.5(M) to M20x1.5(F), with plug	316 SS	6-8 mm	IP67
3	Flameproof adapter M20x1.5(M) to (F), with plug	316 SS	6-12 mm	IP67
4	Flameproof adapter M20x1.5(M) to M20x1.5(F), with plug	316 SS	6-12 mm	IP67
5	Flameproof adapter M20x1.5(M) to M20x1.5(F), with plug	316 SS	6-12 mm	IP67

Ordering Information — Model Code Structure

The FMPT-7000 model code is structured as follows. Please specify all parameters when placing an order. Items marked (Note) have extended lead times. Factory calibration range (LRL-URL) and display unit must be stated in the purchase contract.

Position	Field	Key Options
1	Hazardous Area	See Hazardous Area order-code table, page 6 (0=Non-hazardous; 1-5=China; A/B/C=CSA; E/F/G=ATEX; J/K/L=IECEX; M=EAC)
2	Output Signal	H=4-20 mA HART; B=4-20 mA HART + Bluetooth
3	Case	S=316L SS M20; U=316L SS NPT; P=Aluminium M20; N=Aluminium NPT
4	Waterproof/Explosion-proof Connector	See connector code table above (0=None; A=Flameproof plug; 1=Waterproof; 2-5=Non-FP/FP adapters)
5	Display	N=Without LCD; L=With LCD
6	Range (GP or AP)	GP: 2=0.4 bar; 3=2.5 bar; 4=10 bar; 5=30 bar; 6=100 bar; 7=400 bar. AP: C=0.1 bar; D=0.2 bar; 2=0.4 bar; 3-6=2.5 to 100 bar
7	Sensor Structure	T=Single flange diaphragm seal
8	Wetted Parts	A=316L; B=HC-276; C=Tantalum; E=Ceramic
9	Process Connection	M=M20x1.5; G=G1/2; A=1/2-14NPT(M); N=1/2-14NPT(F); H=Flange (non-diaphragm)
10	Oil Filling	N=No fill (ceramic only); S=Silicone; L=Low-temp silicone; D/E=Fluorocarbon 1/2
/Options	Certificates	/CS1=CCS; /CS2=DNV; /CS3=BV; /CS4=ABS; /CS5=LR; /CS6=KR; /CS7=NK; /CS8=RS
/Options	Special Features	/H7=HART7; /ST=Damping; /J1=Gold diaphragm; /CL1=Oil-free; /G7,/G8=Mounting bracket; /WH,/WL=Fault alarm level
/Options	Other	/Q1=Verification report; /PT=ID plate; /VT=Integrated valve manifold; /QS1-6=Leakage test; /HAC=High accuracy; /PAC,/PBY=Metrological approval (Russia/Belarus); /LE,/LR=Language (English/Russian); /XM=Project-specific delivery
/Options	Warranty	/Y2=2-year; /Y3=3-year; /Y5=5-year warranty

Example: FMPT-7000-GP-0HS1L-4TAMS/G7/Y3 [CAL: 0 to 10 bar, bar display]

Dimensional Reference (mm)

Parameter	With LCD Display	Without LCD Display
Overall Height (male electrical entry)	146.5 mm	126.5 mm
Overall Height (female electrical entry)	147.5 mm	127.5 mm
Overall Width (across electrical entries)	114 mm	114 mm
Depth (with LCD)	140 mm	129 mm
Pressure Port	Per process connection code (M/G/A/N/H)	

Note: Actual dimensions vary with process adapter and mounting bracket selection. Cable gland is supplied as an accessory to be installed by the user.

Quick Reference — Common Configurations

Worked examples for reference only; confirm every field against the full model code table before ordering.

Configuration	Order Code	Range
Standard, non-hazardous, HART, LCD	FMPT-7000-GP-0HS1L-4TAMS	0-10 bar
ATEX flameproof, HART, LCD	FMPT-7000-GP-EHS1L-4TAMS	0-10 bar
CSA intrinsically safe, HART + Bluetooth, LCD	FMPT-7000-GP-BBS1L-4TAMS	0-10 bar
Absolute pressure, ceramic diaphragm, non-hazardous	FMPT-7000-AP-0HS1L-CTEGN	0-0.1 bar
High-pressure GP, Hastelloy wetted parts, ATEX	FMPT-7000-GP-EHS1L-6TBMS	0-100 bar
Marine-certified GP, DNV, HART, aluminium case	FMPT-7000-GP-0HP1L-4TAMS/CS2	0-10 bar